BIG DATA

NATURAL LANGUAGE PROCESSING

GOALS OF THIS PRESENTATION

- At the end of this presentation, you should be able to
  - understand the acronym ’OCR’;
  - explain what purposes NLP can be used for;
  - discuss the basic NLP steps and their purpose;
  - explain what a TF-IDF matrix is and why we create it;
  - explain what word embeddings are.

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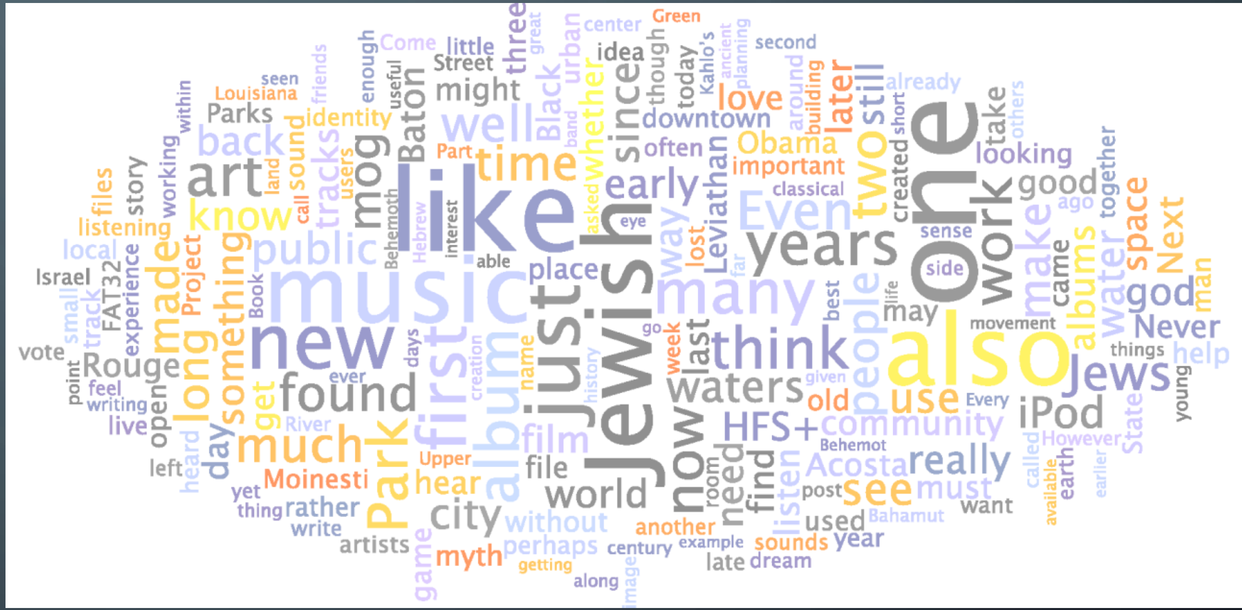
APPLICATIONS

Benefits of NLP

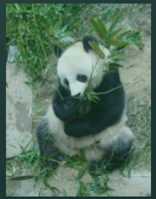


WHAT IS NLP?

- Extracting meaning from text programmatically.
  - 'Natural language': English, Hindi, Chinese, Finnish, Klingon..
- Natural language is tricky
  - It is often ambiguous
- People know the meaning from context
  - This is hard for machines

Eats, shoots and leaves

Eats shoots and leaves

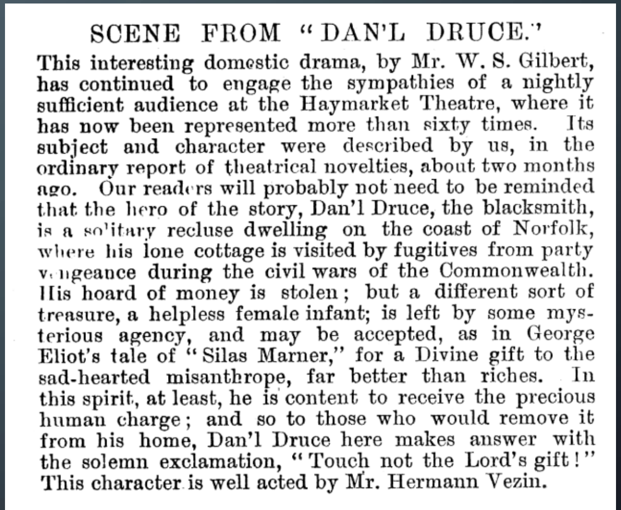


WHAT IS SO GREAT ABOUT NLP?

- Being able to give voice commands to devices changes everything.
  - No need to type or even be near a computer.
    - Ask about details in an instruction manual while building things.
  - Interact with any device via voice control.
    - Adjust the air conditioner on the fly.
  - Enjoy instant translation into any language.
- It's not all about voice, though.

APPLICATIONS OF NLP

- Classification of texts
  - Is an e-mail legitimate or spam?
- Information retrieval
  - Can a computer answer a question by searching documents or web pages?
- Machine translation
  - Ability to communicate across language barriers.
- Summarisation
  - When there is too much data, summarising the important parts can save time.



SCENE FROM "DAN'L DRUCE."

This interesting domestic drama, by Mr. W. S. Gilbert, has continued to engage the sympathies of a nightly sufficient audience at the Haymarket Theatre, where it has now been represented more than sixty times. Its subject and character were described by us, in the ordinary report of theatrical novelties, about two months ago. Our readers will probably not need to be reminded that the hero of the story, Dan'l Druce, the blacksmith, is a solitary recluse dwelling on the coast of Norfolk, where his lone cottage is visited by fugitives from party vengeance during the civil wars of the Commonwealth. His hoard of money is stolen; but a different sort of treasure, a helpless female infant; is left by some mysterious agency, and may be accepted, as in George Eliot's tale of "Silas Marner," for a Divine gift to the sad-hearted misanthrope, far better than riches. In this spirit, at least, he is content to receive the precious human charge; and so to those who would remove it from his home, Dan'l Druce here makes answer with the solemn exclamation, "Touch not the Lord's gift!" This character is well acted by Mr. Hermann Vezin.

BEFORE NLP

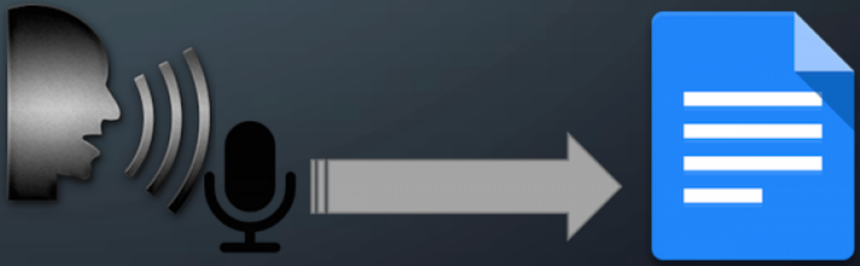
(how to make text)

THE PREPARATION STEP (BEFORE NLP)

- Optical Character Recognition (OCR)
  - Interprets letters in a picture.
  - Needed for scanned documents.

Contract
between C. Longfoot and
S. Barrymore who pledge
not to shoot each other any
more.
x

- Voice (Speech) Recognition
  - Spoken words to text
  - Siri, Alexa



PREPROCESSING

Extracting meaningful parts

NLP - EXAMPLE

1. They took to the hills & were soon out of sight.
2. All is well that ends well.
3. I didn't like the way they were looking at me.

NLP - TOKENISATION

1. ”they” ”took” ”to” ”the” ”hills” ”were” ”soon” ”out” ”of” ”sight”
2. ”all” ”is” ”well” ”that” ”ends” ”well”
3. ”i” ”didn’t” ”like” ”the” ”way” ”they” ”were” ”looking” ”at” ”me”

Removed: & . . . (ampersand and three fullstops)

NLP PROCESSING – REMOVING STOP WORDS

”stop words” = words that do not have much meaning by themselves

1. ~~”they”~~ ”took” ~~”to”~~ ~~”the”~~ ”hills” ~~”were”~~ ”soon” ~~”out”~~ ~~”of”~~ ”sight”
2. ~~”all”~~ ~~”is”~~ ”well” ~~”that”~~ ”ends” ”well”
3. ~~”i”~~ ~~”didn’t”~~ ”like” ~~”the”~~ ”way” ~~”they”~~ ~~”were”~~ ”looking” ~~”at”~~ ~~”me”~~

1. ”took” ”hills” ”soon” ”sight”
2. ”well” ”ends” ”well”
3. ”like” ”way” ”looking”

NLP – STEMMING (LEMMATISATION)

1. ”took” ”hills” ”soon” ”sight”
2. ”well” ”ends” ”well”
3. ”like” ”way” ”looking”

same meaning:
to look – looks – looking

1. ”took” ”hill” ”soon” ”sight”
2. ”well” ”end” ”well”
3. ”like” ”way” ”look”

but:
someone’s looks ≠ someone looks

BAG OF WORDS

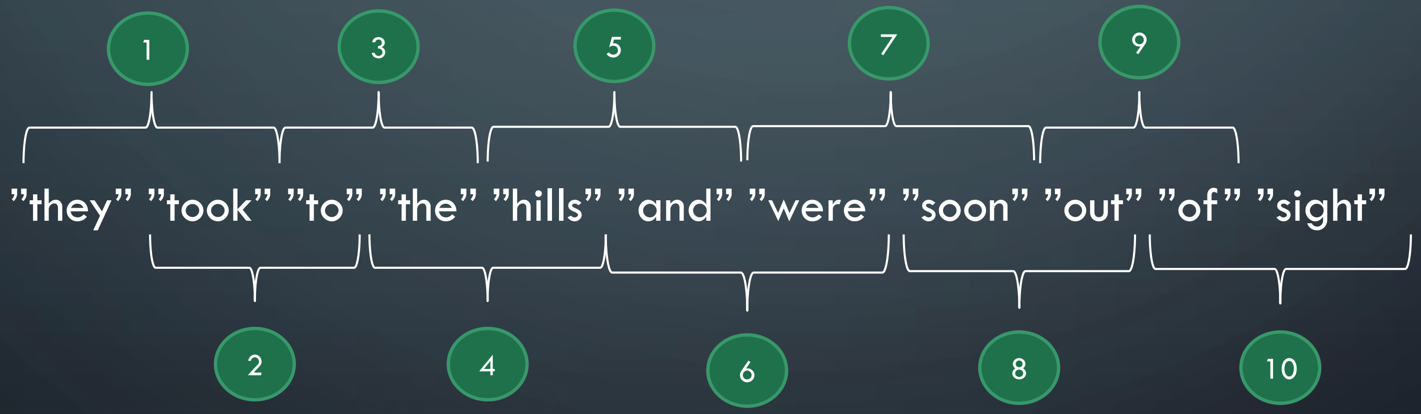
Three bags of words

1. ”took” ”hill” ”soon” ”sight”
2. ”well” ”end” ”well”
3. ”like” ”way” ”look”

Result

TOKENISATION - NGRAMS

- Bag of Words loses the word order.
  - To fix this, we can extract ngrams.
  - n should not be bigger than 5 or 6 (curse of dimensionality)



NLP – 2GRAMS

1. "we_took" "took_to" "to_the" "the_hill" "hill_and" "and_were" "were_soon" "soon_out" "out_of" "of_sight"

original: &

2. "all_is" "is_well" "well_that" "that_end" "end_well "
3. "i_didn't" "didn't_like" "like_the" "the_way" "way_they" "they_were" "were_look" "look_at" "at_me"

meaning preserved!

two stop words

NLP – 3GRAMS

1. "we_took_to" "took_to_the" "to_the_hill" "the_hill_and" "hill_and_were" "and_were_soon" "were_soon_out" "soon_out_of" "out_of_sight“
2. "all_is_well" "is_well_that" "well_that_end" "that_end_well"
3. "i_didn't_like" "didn't_like_the" "like_the_way" "the_way_they" "way_they_were" "they_were_look" "were_look_at" "look_at_me"

NLP – POS TAGS

POS
= parts of speech

Word	POS	POS explained
They	PRP	personal pronoun
took	VBD	verb, past tense
to	TO	to -> the preposition "to"
the	DT	determiner
hills	NNS	noun, plural
and	CC	coordinating conjunction
were	VBD	verb, past tense
soon	RB	adverb
out	IN	preposition or subordinating conjunction
of	IN	preposition or subordinating conjunction
sight	NN	noun, singular or mass

NLP

DFM

TF-IDF

Word embedding

DOCUMENT FREQUENCY MATRIX (FROM BAG OF WORDS)

B
o
W

1. ”took” ”hill” ”soon” ”sight”
2. ”well” ”end” ”well”
3. ”like” ”way” ”look”

Doc	took	hill	soon	sight	well	end	like	way	look
1	1	1	1	1	0	0	0	0	0
2	0	0	0	0	2	1	0	0	0
3	0	0	0	0	0	0	1	1	1

D
F
M

DOCUMENT FREQUENCY MATRIX BIGRAMS

B o W

1. "they_took" "took_to" "to_the" "the_hill" "hill_were" "were_soon" "soon_out" "out_of" "of_sight“
2. "all_is" "is_well" "well_that" "that_end" "end_well"
3. "i_didn't" "didn't_like" "like_the" "the_way" "way_they" "they_were" "were_look" "look_at" "at_me"

Doc	they_took	took_to	to_the	the_hill	hill_were	were_ soon	soon_out	out_of	of_sight
1	1	1	1	1	1	1	1	1	1
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0

D F M

DFM 23 columns instead of 10

DFM AND MACHINE LEARNING

- RANDOM FOREST, SVM, NEURAL NETWORKS...

classify

correlate

eactxtr

Use Twitter feed to find out who is attending an event:

TAG	Tweet	like	great	boring	last	good	awesome	ew	bad	fine
going	1	1	1	0	1	1	0	0	0	0
not going	2	0	0	1	0	0	0	1	1	0
going	3	0	0	0	0	1	1	0	1	1

Model

like, great, good, awesome, fine → attending

boring, ew, bad → not attending

DRAWBACKS OF DFM

Doc	like	go	jurassic	equine	flu	sabre	
1	2	1	0	1	0	0	
2	3	2	0	0	0	0	Length: 500 words
3	1	2	0	0	5	0	
4	6	4	1	0	0	1	Length: 50,000 words
5	1	3	0	0	2	0	

3/500 is more than 6/50,000

TERM FREQUENCY – TF

	i=1	*i=2*	*i=3*	*i=4*	*i=5*	*etc.*	
Doc	**like**	**go**	**jurassic**	**equine**	**flu**	**sabre**	**Doc length**
1	2	1	0	1	0	0	100
2	3	2	0	0	0	0	500
3	1	2	0	0	5	0	4,000
4	6	4	1	0	0	1	50,000

j

$$TF(t_x, d_j) = \frac{freq(t_x, d_j)}{\sum_i^n freq(t_i, d_j)}$$

$$TF(t_1, d_1) = \frac{freq(t_1,d_1)}{\sum_i^n freq(t_i,d_1)} = \frac{2}{2+1+0+1+\cdots}$$

$$= \frac{2}{100}$$

t is term (e.g. 'like')

d is a document

i is an index (e.g. 'like' is 1, go is 2)

j is an index (e.g. doc 1 is 1, doc 2 is 2)

TERM FREQUENCY – TF

Result

Doc	like	go	jurassic	equine	flu	sabre	Doc length
1	2	1	0	1	0	0	100
2	3	2	0	0	0	0	500
3	1	2	0	0	5	0	4,000
4	6	4	1	0	0	1	50,000

Doc	like	go	jurassic	equine	flu	sabre	Doc length
1	.02	.01	0	.01	0	0	100
2	.006	.004	0	0	0	0	500
3	.00025	.0005	0	0	.00125	0	4,000
4	.00012	.00008	.00002	0	0	.00002	50,000

INVERSE DOCUMENT FREQUENCY – IDF

	i=1	*i=2*	*i=3*	*i=4*	*i=5*	*etc.*	
Doc	**like**	**go**	**jurassic**	**equine**	**flu**	**sabre**	**Doc length**
1	2	1	0	1	0	0	100
2	3	2	0	0	0	0	500
3	1	2	0	0	5	0	4,000
4	6	4	1	0	0	1	50,000
Total	4	4	1	1	1	1	N/A

Assuming 150 documents

$$IDF(t_i) = log \frac{|D|}{|t_i \in D|}$$

$$IDF(t_1) = log \frac{150}{4}$$

t is term (e.g. 'like')

d is a document, D is the set of all documents

i is an index (e.g. 'like' is 1, go is 2)

i is an index (e.g. 'like' is 1, go is 2)

INVERSE DOCUMENT FREQUENCY – IDF

Result

	like	go	jurassic	equine	flu	sabre
Occurrence	4	4	1	1	1	1

	like	go	jurassic	equine	flu	sabre
IDF	1.57	1.57	2.18	2.18	2.18	2.18

TF-IDF

Result

$$TF - IDF(t_i, d_j) = TF(t_i, d_j) * IDF(t_i)$$

- Quantifying the significance within document and
- significance across documents
- in an almost normalised way

TF-IDF

Result

Doc	like	go	jurassic	equine	flu	sabre
1	2	1	0	1	0	0
2	3	2	0	0	0	0
3	1	2	0	0	5	0
4	6	4	1	0	0	1

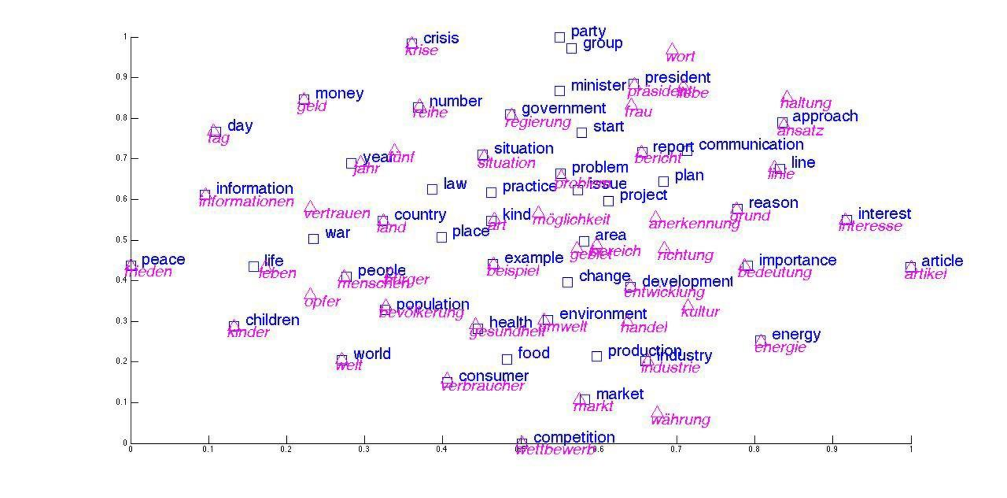
DFM

Doc	like	go	jurassic	equine	flu	sabre
1	.03	.0157	0	.0218	0	0
2	.009	.006	0	0	0	0
3	.0004	.0008	0	0	.0027	0
4	.0002	.0001	.00004	0	0	.00004

TF-IDF

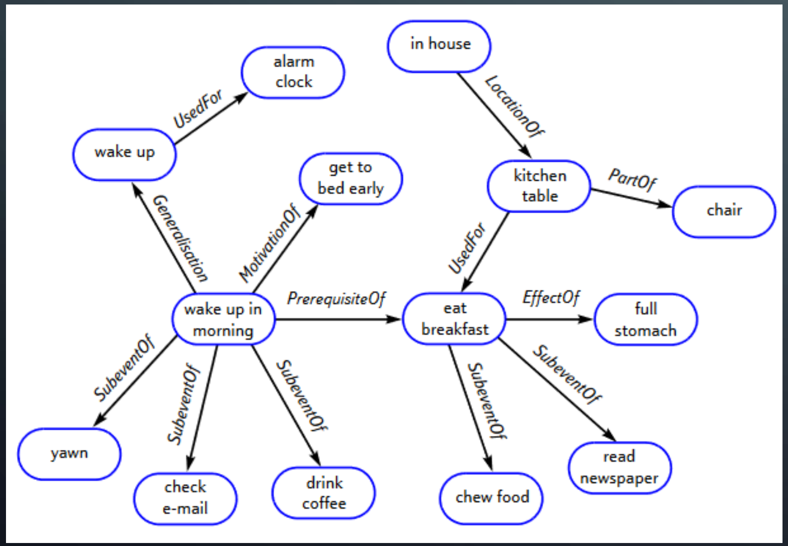
WORD EMBEDDINGS

- Word2vec
- Glove
- fastText
- Gensim

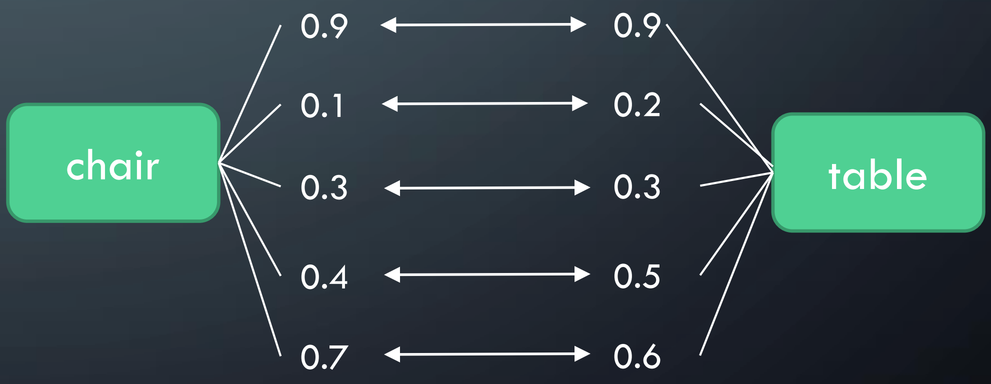


SEMANTIC NETWORK - CONCEPTNET

- A set of words and phrases, each with a vector of 600 numbers that express its meaning.
  - 700 000 ’commonsense’ sentences automatically analysed



http://agents.media.mit.edu/projects/commonsense/ConceptNet-BTTJ.pdf



We don’t know what the numbers mean

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SUMMARY

- Computers become a lot more useful if they understand human language.
  - Machine translations, question answering, voice control, summarisation..
- Computers have to extract meaning from sentences. To be efficient, they have to separate the irrelevant from the relevant.
  - Tokenisation, Bag of words, POS tagging, context determination are all techniques to extract relevant text for easier interpretation
  - TF-IDF is the final step from words to numbers, measuring the occurrence of words in documents
- After these NLP preprocessing steps, analysis and interpretation are possible:
  - Machine learning methods using e.g. Bag of Words or TF-IDF;
  - Neural networks using word embeddings, and many others.